# Differentiation and Rates of Change

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Differentiation & Rates of Change Practice Exercises**

Find the exact area of the region enclosed between the curve $y = 4x - x^2$ and the line $y = x$.

Find points of intersection: $4x - x^2 = x \Rightarrow 3x - x^2 = 0 \Rightarrow x(3 - x) = 0$ The curves intersect at $x = 0$ and $x = 3$. Between $0$ and $3$, the curve $y = 4x - x^2$ lies above the line $y = x$. The enclosed area is: $A = \int_0^3 [(4x - x^2) - x]\,dx = \int_0^3 (3x - x^2)\,dx$ Integrate: $A = \left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^3 = \left(\frac{3(9)}{2} - \frac{27}{3}\right) - 0 = \frac{27}{2} - 9 = \frac{27-18}{2} = \frac{9}{2} = 4.5$

**2** **[Foundational]** • **Target: Differentiation & Rates of Change Practice Exercises**

A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

(a) Find $\frac{dy}{dx}$: $\frac{dy}{dx} = 6x^2 - 18x + 12$ Set $\frac{dy}{dx} = 0$: $6(x^2 - 3x + 2) = 0 \Rightarrow 6(x - 1)(x - 2) = 0$, giving $x = 1$ and $x = 2$. When $x = 1$: $y = 2(1) - 9(1) + 12(1) + 1 = 6$. Point is $(1, 6)$. When $x = 2$: $y = 2(8) - 9(4) + 12(2) + 1 = 16 - 36 + 24 + 1 = 5$. Point is $(2, 5)$. (b) Second derivative: $\frac{d^2y}{dx^2} = 12x - 18$ At $x = 1$: $\frac{d^2y}{dx^2} = 12(1) - 18 = -60$, so $(2, 5)$ is a local minimum.

**3** **[Foundational]** • **Target: Differentiation & Rates of Change Practice Exercises**

A curve is defined parametrically by $x = t^2 + 1$, $y = 2t^3 - 3t$. Find $\frac{dy}{dx}$ in terms of $t$ and the value at $t = 1$.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 6t^2 - 3$. $\frac{dy}{dx} = \frac{6t^2 - 3}{2t}$. At $t = 1$: $\frac{3}{2}$.

**4** **[Foundational]** • **Target: Differentiation & Rates of Change Practice Exercises**

In triangle $OAB$, $\overrightarrow{OA} = 3\mathbf{a}$ and $\overrightarrow{OB} = 6\mathbf{b}$. Point $P$ lies on $AB$ such that $AP : PB = 1 : 2$. Find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ in its simplest form.

Step 1: Express $\overrightarrow{AB}$: $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -3\mathbf{a} + 6\mathbf{b}$ Step 2: Find $\overrightarrow{AP}$: $\overrightarrow{AP} = \frac{1}{3}\overrightarrow{AB} = \frac{1}{3}(-3\mathbf{a} + 6\mathbf{b}) = -\mathbf{a} + 2\mathbf{b}$ Step 3: Find $\overrightarrow{OP}$: $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 3\mathbf{a} + (-\mathbf{a} + 2\mathbf{b}) = 2\mathbf{a} + 2\mathbf{b}$

**5** **[Foundational]** • **Target: Differentiation & Rates of Change Practice Exercises**

An ellipse has vertices at $(1, 6)$ and $(1, -4)$, and a focus at $(1, 4)$. Determine its standard equation and compute its eccentricity.

The vertices share $x = 1$, so the major axis is vertical. Center is the midpoint of the vertices: $(h, k) = \left(1, \frac{6 + (-4)}{2}\right) = (1, 1)$. Semi-major axis: $a = 6 - 1 = 5$ (so $a^2 = 25$). Focal distance from center to $(1, 4)$: $c = 4 - 1 = 3$. For an ellipse, $b^2 = a^2 - c^2 = 25 - 9 = 16 \Rightarrow b = 4$. Standard equation: $\frac{(x-1)^2}{16} + \frac{(y-1)^2}{25} = 1$. Eccentricity: $e = \frac{c}{a} = \frac{3}{5} = 0.6$.

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**6** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

The curve $y = x^2 - 6x + 5$ can be mapped to $y = x^2$ by a sequence of two geometric transformations. Express the equation in vertex form, state the vertex, and describe fully each transformation.

Complete the square: $y = (x-3)^2 - 9 + 5 = (x-3)^2 - 4$. The vertex of the parabola is $(3,\ -4)$. To map $y = (x-3)^2 - 4$ back to $y = x^2$, we must translate the graph 3 units to the left and 4 units upwards. In vector notation, this is a single translation by $(-3; 4)$, or two sequential transformations: (1) Translation by vector $(-3; 0)$ (3 units left), and (2) Translation by vector $(0; 4)$ (4 units up).

**7** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Eliminate the parameter to determine the Cartesian equation of the curve $x = 1 + 2\sec t$, $y = -2 + 3\tan t$. State the shape, center, and vertices.

Isolate the trigonometric terms: $\sec t = \frac{x-1}{2}, \quad \tan t = \frac{y+2}{3}$. Apply the fundamental Pythagorean identity $\sec^2 t - \tan^2 t = 1$: $\left(\frac{x-1}{2}\right)^2 - \left(\frac{y+2}{3}\right)^2 = 1 \Rightarrow \frac{(x-1)^2}{4} - \frac{(y+2)^2}{9} = 1$. This is a horizontal hyperbola with:Center: $(h, k) = (1,\ -2)$Transverse semi-axis: $a = 2$Vertices: $(h \pm a, k) = (1 \pm 2,\ -2) = (-1,\ -2)$ and $(3,\ -2)$

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

A car of mass $1200$ kg ascends a hill of inclination $\alpha$, where $\sin\alpha = \frac{1}{20}$, against a constant resistance to motion of $500$ N. Taking $g = 9.8$ m/s$^2$: (a) find the driving force required to travel at a steady speed of $25$ m/s, (b) calculate the power developed by the engine in kilowatts.

(a) Component of weight down the slope: $W_{\parallel} = mg\sin\alpha = 1200(9.8)(\frac{1}{20}) = 60(9.8) = 588$ N Since the car travels at steady speed (acceleration $a = 0$), the engine driving force balances all resistance: $F_{\text{drive}} = \text{Resistance} + mg\sin\alpha = 500 + 588 = 1088$ N (b) Power output: $P = F_{\text{drive}} \times v = 1088 \times 25 = 27\,200$ W $= 27.2$ kW

**9** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

On an Argand diagram, the point $P$ represents the complex number $z$ such that $|z - (2 + 3i)| \leq 2$. Find the greatest possible value of $|z|$ for points in this region.

The locus $|z - (2 + 3i)| \leq 2$ represents a circular disc of radius $R = 2$ with center at $C(2, 3)$. The distance from the origin $O(0, 0)$ to the center $C$ is: $|C| = \sqrt{2^2 + 3^2} = \sqrt{4 + 9} = \sqrt{13} \approx 3.6055$ The point on the disc farthest from the origin lies on the line passing through $O$ and $C$, on the far boundary of the circle: $|z|_{\max} = |C| + R = \sqrt{13} + 2 \approx 3.606 + 2 = 5.61$ (3 s. f. )

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**10** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Events $A$ and $B$ are such that $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cup B) = 0.8$. Find $P(A \cap B)$, calculate $P(A|B)$, and determine whether $A$ and $B$ are independent.

1. Use the addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow 0.8 = 0.6 + 0.5 - P(A \cap B) \Rightarrow P(A \cap B) = 1.1 - 0.8 = 0.3$. 2. Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.6$ 3. Check for independence: $P(A \cap B) = 0.3$ and $P(A) \times P(B) = 0.6 \times 0.5 = 0.30$. Since $P(A \cap B) = P(A)P(B)$ (or equivalently $P(A|B) = P(A) = 0.6$), events $A$ and $B$ are independent.

**11** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the equation of the perpendicular bisector of the line segment joining $A(-2, 5)$ and $B(4, -3)$. Give your answer in the form $ax + by + c = 0$, where $a, b, c$ are integers.

1. Midpoint of $AB$: $M = \left(\frac{-2+4}{2}, \frac{5-3}{2}\right) = (1, 1)$. 2. Gradient of $AB$: $m_{AB} = \frac{-3-5}{4-(-2)} = \frac{-8}{6} = -\frac{4}{3}$. 3. Gradient of perpendicular bisector: $m_{\perp} = -\frac{1}{-4/3} = \frac{3}{4}$. 4. Equation of line passing through $M(1, 1)$: $y - 1 = \frac{3}{4}(x - 1) \Rightarrow 4(y - 1) = 3(x - 1) \Rightarrow 4y - 4 = 3x - 3$ Rearrange to standard integer form: $3x - 4y + 1 = 0$.

**12** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

The random variable $X \sim N(\mu, \sigma^2)$ satisfies $P(X < 65) = 0.8413$ and $P(X < 45) = 0.0228$. Find the values of $\mu$ and $\sigma$.

Standardising: $\Phi\left(\frac{65-\mu}{\sigma}\right) = 0.8413 \Rightarrow \frac{65-\mu}{\sigma} = 1.0$, so $65 - \mu = \sigma$. For the lower tail: $P(X < 45) = 0.0228 \Rightarrow \Phi\left(\frac{45-\mu}{\sigma}\right) = 0.0228 \Rightarrow \frac{45-\mu}{\sigma} = -2.0$, so $\mu - 45 = 2\sigma$. Adding the two equations: $65 - 45 = 3\sigma \Rightarrow 20 = 3\sigma \Rightarrow \sigma = 6.67$. Then $\mu = 65 - 6.67 = 58.33$.

**13** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

A solid toy is formed by attaching a cone of radius 4 cm and height 9 cm to the flat face of a solid hemisphere of radius 4 cm. Calculate the total volume of the toy. Give your answer correct to 3 significant figures.

Step 1: Volume of cone: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(4^2)(9) = 48\pi \text{ cm}^3$ Step 2: Volume of hemisphere: $V_{\text{hemi}} = \frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2}{3}\pi(4^3) = \frac{128}{3}\pi \text{ cm}^3$ Step 3: Total volume: $V_{\text{total}} = 48\pi + \frac{128}{3}\pi = \frac{272}{3}\pi \approx 284.838\ldots \approx 285 \text{ cm}^3$ (3 s. f. )

**14** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the angle $\theta$ between the vectors $\mathbf{u} = \langle 1, \sqrt{3} \rangle$ and $\mathbf{v} = \langle 2, 0 \rangle$ in degrees.

Step 1: Compute dot product and magnitudes: $\mathbf{u} \cdot \mathbf{v} = (1)(2) + (\sqrt{3})(0) = 2$. $||\mathbf{u}|| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2, \quad ||\mathbf{v}|| = \sqrt{2^2 + 0^2} = 2.$ Step 2: Solve for $\cos\theta$: $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||} = \frac{2}{(2)(2)} = \frac{2}{4} = \frac{1}{2} \Rightarrow \theta = \arccos(1/2) = 60^\circ$.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Differentiation & Rates of Change Practice Exercises** • **Cross-Lesson: 8.01 Integration and Kinematics**

A student calculates $\int_1^3 \frac{4}{2x+1}\,dx$ by rounding $\ln 7 \approx 1.95$ and $\ln 3 \approx 1.10$ at the start, obtaining $2(1.95 - 1.10) = 1.70$. State the exact value of the integral and calculate the absolute error introduced by premature rounding.

Integrate exactly: $\int_1^3 \frac{4}{2x+1}\,dx = [2\ln(2x+1)]_1^3 = 2\ln 7 - 2\ln 3 = 2\ln\left(\frac{7}{3}\right) \approx 1.6946\ldots$ The exact value is $2\ln\left(\frac{7}{3}\right)$. The premature rounding gives $1.70$, yielding an absolute error of: $|1.70 - 1.6946| \approx 0.0054$ On Cambridge Paper 3, this premature rounding costs the final accuracy mark (A1) because the 3 s.f. answer is 1.69, not 1.70.

**16** **[Challenging]** • **Target: Differentiation & Rates of Change Practice Exercises** • **Cross-Lesson: 1.01 Functions, Modulus and Quadratics**

The masses, $x$ grams, of 50 packets of tea gave $\sum x = 5090$ and $\sum x^2 = 518380$. (a) Calculate unbiased estimates of the population mean and variance. (b) Calculate a 95% confidence interval for the population mean.

(a) Sample mean: $\bar{x} = \frac{5090}{50} = 101.8$ g Unbiased sample variance: $s^2 = \frac{1}{50-1}(518380 - 50(101.8)^2) = \frac{1}{49}(518380 - 518162) = \frac{218}{49} \approx 4.449$ g² (b) Standard error: $\text{SE} = \sqrt{\frac{s^2}{n}} = \sqrt{\frac{4.449}{50}} \approx 0.2983$. $\bar{x} \pm 1.96(0.2983) = 101.8 \pm 0.585 = (101.2,\ 102.4)$ g (1 d. p.)

**17** **[Challenging]** • **Target: Differentiation & Rates of Change Practice Exercises** • **Cross-Lesson: 2.01 Simultaneous Equations and Inequalities**

In an arithmetic progression, the 3rd term is $14$ and the 7th term is $34$. (a) Find the first term $a$ and the common difference $d$. (b) Find the sum of the first $25$ terms.

(a) Use $u_n = a + (n-1)d$: $u_3 = a + 2d = 14$ $u_7 = a + 6d = 34$ Subtract the first equation from the second: $4d = 20 \Rightarrow d = 5$. Substitute back: $a + 2(5) = 14 \Rightarrow a = 4$. (b) Use the sum formula $S_n = \frac{n}{2}(2a + (n-1)d)$ with $n = 25$: $S_{25} = \frac{25}{2}(2(4) + 24(5)) = \frac{25}{2}(8 + 120) = \frac{25}{2}(128) = 25 \times 64 = 1600$

**18** **[Challenging]** • **Target: Differentiation & Rates of Change Practice Exercises** • **Cross-Lesson: 3.01 Indices, Surds and Polynomials**

Find the vertex, focus, and directrix of the parabola given by the general equation $y^2 + 8x - 4y + 28 = 0$.

Complete the square on the $y$-terms: $y^2 - 4y = -8x - 28 \Rightarrow (y^2 - 4y + 4) = -8x - 28 + 4$. $(y-2)^2 = -8x - 24 = -8(x+3)$. This matches $(y-k)^2 = 4p(x-h)$ with:Vertex: $(h,k) = (-3, 2)$ Focal parameter: $4p = -8 \Rightarrow p = -2$ (opens horizontally to the left)Focus: $(h+p, k) = (-3 + (-2), 2) = (-5, 2)$Directrix: $x = h - p = -3 - (-2) \Rightarrow x = -1$

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**19** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A 3 kg particle on a rough surface, $\mu = 0.4$. Horizontal force $F$ N. (a) Find the limiting friction. (b) Acceleration when $F = 20$ N. (c) Time to travel 5 m from rest.

(a) $F_f = 0.4 \times 3 \times 9.8 = 11.76$ N. (b) $a = \frac{20 - 11.76}{3} \approx 2.75$ m s$^{-2}$. (c) $t = \sqrt{10/2.75} \approx 1.91$ s. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

**20** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

A surveyor measures distances from an observation point $C$ to two landmarks $A$ and $B$: $AC = 140$ m and $BC = 210$ m, with included angle $\angle ACB = 78^\circ$. Find the distance $AB$ between the landmarks to the nearest tenth of a meter, and compute the area of $\Delta ABC$ to the nearest square meter.

Apply the Law of Cosines to find side $c = AB$: $c^2 = a^2 + b^2 - 2ab\cos C = 210^2 + 140^2 - 2(210)(140)\cos 78^\circ$. $c^2 = 44100 + 19600 - 58800(0.2079) = 63700 - 12225 = 51475$. $c = \sqrt{51475} \approx 226.9$ m. Compute the area: Area $= \frac{1}{2}ab\sin C = \frac{1}{2}(210)(140)\sin 78^\circ = 14700(0.9781) \approx 14379$ m$^2$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to \( x \): \( y = 3x^{3} - 5x + 7 \).

**a.** \( \frac{dy}{dx} = 3x^{2} - 5 \)

**b.** \( \frac{dy}{dx} = 9x^{2} \)

**c.** \( \frac{dy}{dx} = 9x^{3} - 5 \)

**d.** **\( \frac{dy}{dx} = 9x^{2} - 5 \)** ✓

Apply the power rule \( \frac{d}{dx}[x^n] = n x^{n-1} \): \( \frac{dy}{dx} = 3(3)x^{2} - 5 = 9x^{2} - 5 \).

**22** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve is given by the equation $y = x^3 - 3x^2 + 2x$. Find the gradient of the curve at $x = 2$.

**a.** 1

**b.** **-2** ✓

**c.** 2

**d.** 0

To find the gradient of the curve, we need to find the first derivative of the equation with respect to $x$. The derivative of $y = x^3 - 3x^2 + 2x$ is $\frac{dy}{dx} = 3x^2 - 6x + 2$. To find the gradient at $x = 2$, substitute $x = 2$ into the derivative: $\frac{dy}{dx}|_{x=2} = 3(2)^2 - 6(2) + 2 = 3(4) - 12 + 2 = 12 - 12 + 2 = 2$.

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**23** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A straight line passing through the origin is tangent to both the curve $C_1$ with equation $y = e^{kx}$, where $k$ is a positive constant, and the curve $C_2$ with equation $y = \ln x$. Determine the exact distance between the two points of contact.

**a.** $(e-1)\sqrt{e^2+1}$ ✓

**b.** $e\sqrt{e^2+1}$

**c.** $(e-1)\sqrt{e^2-1}$

**d.** $\sqrt{e^4-e^2+1}$

We synthesize differentiation with coordinate geometry to find the two points of contact. **Step 1: Point of tangency on $C_2 : y = \ln x$** Let $A(x_1, y_1)$ be the point of contact on $C_2$, so $y_1 = \ln x_1$. The derivative is $\frac{dy}{dx} = \frac{1}{x}$, so the gradient of the tangent at $A$ is $\frac{1}{x_1}$. Since the tangent line passes through the origin $(0, 0)$, its gradient is also: $m = \frac{y_1 - 0}{x_1 - 0} = \frac{\ln x_1}{x_1}$ Equating the two expressions for the gradient.

**24** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

The curve $C$ has equation $y = \frac{ax+6}{x^2+3}$ where $a$ is a constant. Given that $C$ has a stationary point at $x = 1$, find the coordinates of the other stationary point of $C$.

**a.** $(-3, -1)$ ✓

**b.** $(-3, 1)$

**c.** $(3, 1)$

**d.** $(-3, -2)$

To find the unknown parameter $a$, differentiate $y$ with respect to $x$ using the quotient rule: $\frac{dy}{dx} = \frac{a(x^2+3)-(ax+6)(2x)}{(x^2+3)^2} = \frac{3a-12x-ax^2}{(x^2+3)^2}$ At a stationary point, $\frac{dy}{dx} = 0$, which requires the numerator to be zero: $3a - 12x - ax^2 = 0$ Given that $x = 1$ is a stationary point: $3a - 12(1) - a(1)^2 = 0 \Rightarrow 2a - 12 = 0 \Rightarrow a = 6$ Substitute $a = 6$ back into the numerator of the derivative.

**25** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?

**a.** 2 ✓

**b.** 6

**c.** $\frac{2}{3}$

**d.** $\ln 3$

To find the gradient of the tangent to the curve $y = \ln(2x^3 + 1)$ at $x = 1$, we differentiate $y$ with respect to $x$ using the chain rule. Let $u = 2x^3 + 1$, so $y = \ln(u)$. Then: $\frac{du}{dx} = 6x^2$ and $\frac{dy}{du} = \frac{1}{u}$ Applying the chain rule: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{6x^2}{2x^3+1}$ Now evaluate the derivative at $x = 1$.

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**26** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve has equation $y = x^2e^{3x}$. Find the gradient of the curve at the point where $x = 1$.

**a.** $5e^3$ ✓

**b.** $6e^3$

**c.** $3e^3$

**d.** $2e^3$

To find the gradient of the curve, differentiate $y = x^2e^{3x}$ with respect to $x$ using the product rule: $\frac{d}{dx}[uv] = u\frac{dv}{dx} + v\frac{du}{dx}$ Let $u = x^2$ and $v = e^{3x}$. Then $\frac{du}{dx} = 2x$ and, by the chain rule, $\frac{dv}{dx} = 3e^{3x}$. Applying the product rule: $\frac{dy}{dx} = x^2(3e^{3x}) + 2x(e^{3x}) = (3x^2 + 2x)e^{3x}$ Substitute $x = 1$.

**27** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve satisfies $x^2 + y^2 = 25$. Find $\frac{dy}{dx}$ at the point $(3, -4)$.

**a.** $-\frac{3}{4}$

**b.** $\frac{3}{4}$ ✓

**c.** $\frac{4}{3}$

**d.** $-\frac{4}{3}$

Differentiating implicitly: $2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$. At $(3, -4)$: $\frac{3}{4}$.

**28** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A curve is defined parametrically by $x = 2t^2$, $y = 3t^3$. Find $\frac{dy}{dx}$ at $t = 1$.

**a.** $\frac{7}{3}$

**b.** $\frac{9}{4}$ ✓

**c.** $-\frac{9}{4}$

**d.** $\frac{4}{9}$

$\frac{dx}{dt} = 4t$, $\frac{dy}{dt} = 9t^2$. $\frac{dy}{dx} = \frac{9t^2}{4t} = \frac{9t}{4}$. At $t = 1$: $\frac{9}{4}$.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the gradient of the normal to $y = 3x^2 + 2x$ at $x = -1$.

**a.** -3/4

**b.** -1/4

**c.** 5/4

**d.** **1/4** ✓

Tangent gradient $= \frac{dy}{dx}\Big|_{x=-1} = -4$. Normal gradient $= -\frac{1}{-4} = 1/4$

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**30** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the $x$-coordinates of the stationary points of $y = 3x^3 - 9/2x^2 - 54x$.

**a.** $x = -2$ or $x = -3$

**b.** $x = -2$ or $x = 3$ ✓

**c.** $x = -1$ or $x = 3$

**d.** $x = 2$ or $x = 3$

$\frac{dy}{dx} = 9x^2 - 9x - 54 = 3(3)(x - 3)(x + 2) = 0 \Rightarrow x = -2$ or $x = 3$

**31** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate $y = (x)^2$ and evaluate $\frac{dy}{dx}$ at $x = 0$.

**a.** 4

**b.** 5

**c.** 1

**d.** **0** ✓

$\frac{dy}{dx} = 2 \times 1 \times (x)^1$. At $x = 0$: $(x) = 0$, giving $0$.

**32** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the area enclosed between the curve $y = x$, the $x$-axis, and the lines $x = 0$ and $x = 3$.

**a.** 19/2

**b.** 9

**c.** **9/2** ✓

**d.** 7/2

Area $= \int_0^3 (x)dx = [1/2x^2]_0^3 = 9/2$

**33** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Given $y = 5e^{2x}$, find the coefficient of $e^{2x}$ in $\frac{dy}{dx}$.

**a.** 11

**b.** **10** ✓

**c.** 2

**d.** 7

$\frac{dy}{dx} = 5 \times 2\, e^{2x} = 10e^{2x}$. Coefficient: 10.

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**34** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the change in gravitational potential energy of a body of mass $18$ kg raised through a height of $15$ m. Take $g = 9.8$ m/s$^2$.

**a.** 5292.00

**b.** 2664.00

**c.** **2646.00 ✓**

**d.** 2628.00

$\Delta GPE = mg\Delta h = 18 \times 9.8 \times 15 = 2646.00$ J

**35** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the gradient of the tangent to $y = 3x^2 - 6x$ at $x = 3$.

**a.** 13

**b.** -12

**c.** 11

**d.** **12 ✓**

$\frac{dy}{dx} = 6x - 6$. At $x = 3$: gradient $= 12$

**36** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

Find the $x$-coordinate of the stationary point of $y = 3(x + 1)^2 - 2$.

**a.** 1

**b.** 0

**c.** **-1 ✓**

**d.** -2

$\frac{dy}{dx} = 6(x + 1) = 0 \Rightarrow x = -1$

**37** [Intermediate] • Target: Differentiation & Rates of Change Practice Exercises

In a right-angled triangle $ABC$, the angle $B$ is 90 degrees. If the length of side $AB$ is 8 units and the length of side $BC$ is 6 units, what is the length of the hypotenuse $AC$?

**a.** **10 units ✓**

**b.** 14 units

**c.** 48 units

**d.** 20 units

This is a right-angled triangle, so we can use the Pythagorean theorem: $a^2 + b^2 = c^2$. Here, the sides are $AB = 8$ and $BC = 6$, and the hypotenuse is $AC$. So, $AC^2 = AB^2 + BC^2$. $AC^2 = 8^2 + 6^2$ $AC^2 = 64 + 36$ $AC^2 = 100$ $AC = \sqrt{100}$ $AC = 10$ units. Common mistakes include adding the sides (8 + 6 = 14) or multiplying them (8 * 6 = 48) instead of applying the Pythagorean theorem.

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Set $A$ has 12 elements and set $B$ has 8 elements. If $A \cap Beq\varnothing$, what is the maximum possible number of elements in $A \cup B$?

**a.** **19** ✓

**b.** 20

**c.** 12

**d.** 11

Formula: $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 12 + 8 - n(A \cap B) = 20 - n(A \cap B)$. To maximize the union, we must minimize the intersection. Since $A \cap Beq\varnothing$, the minimum possible value of $n(A \cap B)$ is 1. Thus maximum $n(A \cup B) = 20 - 1 = 19$. [M1 for maximizing union by minimizing intersection to 1, A1 for 19]

**39** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A rectangular prism has a length of 5 cm, a width of 3 cm, and a height of 4 cm. Calculate the volume of the prism.

**a.** 12 $cm^3$

**b.** **60 $cm^3$** ✓

**c.** 47 $cm^3$

**d.** 72 $cm^3$

The volume of a rectangular prism is calculated by the formula: Volume = length × width × height. Given: Length = 5 cm Width = 3 cm Height = 4 cm Volume = 5 cm × 3 cm × 4 cm Volume = 15 $cm^2$ × 4 cm Volume = 60 $cm^3$ Therefore, the volume of the prism is 60 $cm^3$.

**40** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.

**a.** 5

**b.** $x$ ✓

**c.** 6

**d.** 7

The coordinates of A are (2, 1) and the coordinates of C are (5, 5). Using the distance formula, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Length of AC = $\sqrt{(5-2)^2 + (5-1)^2}$ Length of AC = $\sqrt{(3)^2 + (4)^2}$ Length of AC = $\sqrt{9 + 16}$ Length of AC = $\sqrt{25}$ Length of AC = 5.

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**41** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

Universal set $\mathscr{E} = \{x : x \text{ is an integer}, 1 \leq x \leq 20\}$. $P = \{x : x \text{ is a prime number}\}$ and $O = \{x : x \text{ is an odd number}\}$. What is $n(P \cap O')$?

**a.** **1 ✓**

**b.** 2

**c.** 8

**d.** 9

The prime numbers in $\mathscr{E}$ are $\{2, 3, 5, 7, 11, 13, 17, 19\}$. The set $O'$ consists of all even numbers. The only even prime number is 2, so $P \cap O' = \{2\}$, which has exactly 1 element: $n(P \cap O') = 1$. [M1 for identifying even prime 2, A1 for 1]

**42** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

The diagram shows the graph of the quadratic function $y = x^2 - 4x + k$. The graph intersects the x-axis at point P(1, 0) and the y-axis at point Q. Determine the value of $k$.

**a.** **k = 3 ✓**

**b.** k = -3

**c.** k = 1

**d.** k = -1

The graph of the quadratic function $y = x^2 - 4x + k$ passes through the point P(1, 0). Substituting the coordinates of P into the equation, we get: $0 = (1)^2 - 4(1) + k$ $0 = 1 - 4 + k$ $0 = -3 + k$ $k = 3$. Therefore, the value of $k$ is 3.

**43** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The graph of the function $y = f(x)$ is shown below. Find the value of $f(3)$.

**a.** 2

**b.** **0 ✓**

**c.** -2

**d.** 4

The question asks for the value of $f(3)$, which corresponds to the y-coordinate on the graph when the x-coordinate is 3. Looking at the graph, when $x = 3$, the point on the curve has a y-coordinate of 0. Therefore, $f(3) = 0$.

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**44** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

A number line is shown below. What is the value of the number indicated by the arrow?

**a.** -1

**b.** **0** ✓

**c.** 1

**d.** 2

The number line shows integers from -5 to 7. The arrow points directly to the mark labelled '0'. Therefore, the value of the number indicated by the arrow is 0.

**45** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Universal set $\mathscr{E} = \{x : x \text{ is an integer}, 1 \leq x \leq 10\}$. Set $A = \{2, 3, 5, 7\}$. List the elements of $A'$.

**a.** $\{1, 4, 6, 8, 9, 10\}$ ✓

**b.** $\{4, 6, 8, 9, 10\}$

**c.** $\{1, 4, 6, 8, 10\}$

**d.** $\{2, 3, 5, 7\}$

The complement $A'$ consists of all elements in the universal set $\mathscr{E}$ that are NOT in set $A$:
$A' = \{1, 4, 6, 8, 9, 10\}$. [B1]

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